

Radiocarbon dating of Jerusalem's Siloam Dam links climate data and major waterworks

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Supporting Information - Extended Methods

Excavation details

Excavations at the Siloam Pool (license nos. A9518/2023, A9742/2024) were conducted on behalf of the Israel Antiquities Authority directed by Nahshon Szanton, Itamar Berko, and Filip Vukosavović.

Field sampling

The samples were collected by radiocarbon specialists together with the field archaeologists. In the field, an initial search was made to identify the type of organic remains that could be dated from the dam material. Small pieces of charred remains were found, and in addition, tiny fragments of straw were identified in the yellowish mortar. Uncharred preserved straw from mortar is a chronologically accurate dating material, since it is a short-lived plant that was collected close to the construction time and likely was not stored for a long time before use. However, as the straw was present only sporadically and it is impossible to know the state of preservation before pretreatment, three buckets of mortar, equivalent to 30 litres, were collected from several locations.

From the vertical part of the wall, W001, after cleaning several centimeters of the outer layer of the wall, chunks of hard, yellowish mortar were retrieved from between the medium-sized stones of the wall.

The samples from the slanting enforcement wall W114, attached to W001, were collected from the horizontal top part between and under medium-sized stones from the core of the wall, close to its current outer surface, after several centimeters of the mortar were removed from the surface.

Extraction of botanical remains

The extraction of straw and charred remains was conducted in the laboratory, using a binocular (Olympus SZ51) for the initial search for organic remains and a binocular microscope SMZ 800N (Nikon) for further verification and cleaning of straw from the sediment. The outer surfaces of the plaster were carefully scraped clean using a surgical scalpel. Some roots were identified on the outer surfaces or in areas with voids next to stones, so careful cleaning was essential to avoid possible modern contamination if roots were mistakenly included. The plaster was then gradually broken into smaller chunks using a hammer. Each fresh surface was examined by the naked eye and a binocular to thoroughly identify pieces of straw or charred remains embedded in the plaster. Once identified, they were extracted using dental tools and tweezers. The initial sample weight ranged from 19 mg to 6.7 mg for straw samples, while the final quantity of graphite remaining for measurement was between 0.47 and 0.25 mg. These numbers reflect the preservation of the organic component of the straw and our success in separating the straw from the adhered plaster before chemical pretreatment.

Sample preparation for radiocarbon dating and measurement

The charred twigs and uncharred straw samples were treated according to the Acid-Base-Acid (ABA) method described in Boaretto (1). Pieces of straw collected from several chunks in one basket were accurately weighed into pre-baked glass tubes. The samples were first treated with 1N HCl at room temperature for 1 hour to dissolve the carbonate fraction and then treated with 0.1N NaOH at room temperature for 30 minutes to remove humic substances, followed by 1N HCl at 80°C for 1 hour to release the CO₂ absorbed during the base treatment. Each acid or base treatment is followed by rinsing with Nanopure water until pH=7. After drying overnight at 90°C, the pre-treated sample was weighed.

Graphitization was performed using an EA-IRMS-AGE3 system composed of an elemental analyzer (EA, 'vario ISOTOPE SELECT' by Elementar) coupled with an isotope ratio mass spectrometer (isoprime precisiON by Elementar), along with the third generation of the Automated Graphitization Equipment (AGE3, Ionplus) (2). The elemental analyzer measured the carbon content of the sample, which ranged from 16-28% for the straw and 59% for the charred twigs. The relatively low %C in the straw samples was due to the presence of quartz, as detected by FTIR (see below). The $\delta^{13}\text{C}$ values were measured by the IRMS and fell within typical ranges for plants (-22.8 - -25.2‰). The graphite was then pressed into an aluminum cathode using a pneumatic sample press (PSP, Ionplus). The ¹⁴C content was measured at the Dangoor Research Accelerator Mass Spectrometry Laboratory (D-REAMS) at the Weizmann Institute of Science (3). The radiocarbon ages were calibrated using OxCal software version 4.4.4 (4), in accordance with the IntCal20 atmospheric curve (5).

Fourier Transform Infrared spectroscopy (FTIR)

The samples were characterized using FTIR analysis throughout the pretreatment process. A few milligrams were crushed in an agate mortar and pestle. A small amount of KBr was added, mixed,

and pressed under a pressure of 2 tons to form a 7 mm pellet with a Specac manual hydraulic press. The pellet was then measured with a Nicolet iS5 (Thermo) FTIR instrument at 4 cm⁻¹ resolution for 32 scans. As a reference library for the spectra identification, we used the "Kimmel Center for Archaeological Science Infrared Standards Library, Weizmann Institute of Science," freely available at <https://centers.weizmann.ac.il/kimmel-arch/infrared-spectra-library>.

References

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